



Assignment 02 Election 2019 Analysis

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Web and Social Media Analysis



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Initial Processing

We have converted date from 2019-05-18 23:50:47+00:00 to 2019-05-18. We have cleaned the Tweet column (Convert to lowercase, Remove URLs, Remove HTML entities, Remove User Mentions, Remove Newline characters etc. for further analysis.

Define User validity

In social media analysis, User Validity is crucial because a single bot can generate thousands of tweets, skewing your entire Sentiment Analysis outcome. We have identified 355 potential bot/spam accounts with most active user tweeted 1152 times (approx. 8.4 tweets/day). Since the maximum is only 8 tweets/day, no one is behaving like a "high-frequency machine bot" (which usually tweets 100+ times/day). So, we will focus on tweets

Dataset covers 137 days.				
Identified 11 bot/spam accounts using realistic thresholds.				
	User	Total_Tweets	Unique_Tweets	Spam_Score
9607	KKMMB9	40	1	0.975000
20973	Tippu143u	16	1	0.937500
804	AbdulRa99443551	11	1	0.909091
41453	realVishaL	28	5	0.821429
23191	aa_rkt	15	3	0.800000

which are often & repeat the same content (copy-paste bots). We have identified 11 bot/spam accounts as their Total tweet to Unique tweet ratio is low hence their Spam score is high.

Now we will work on Tweets which are repeated and remove them.

142561	RAMANKAIRA	2019-03-06	early morning political move detailed discussi...
142562	RAMANKAIRA	2019-03-06	early morning political move detailed discussi...
142563	thota_J	2019-01-24	cvoter predicts seat share upcoming loksabhael...
142564	thota_J	2019-01-24	cvoter predicts seat share upcoming loksabhael...
142565	thota_J	2019-01-24	cvoter predicts seat share upcoming loksabhael...

Now our data is ready for Further Analysis.

Sentiment Analysis

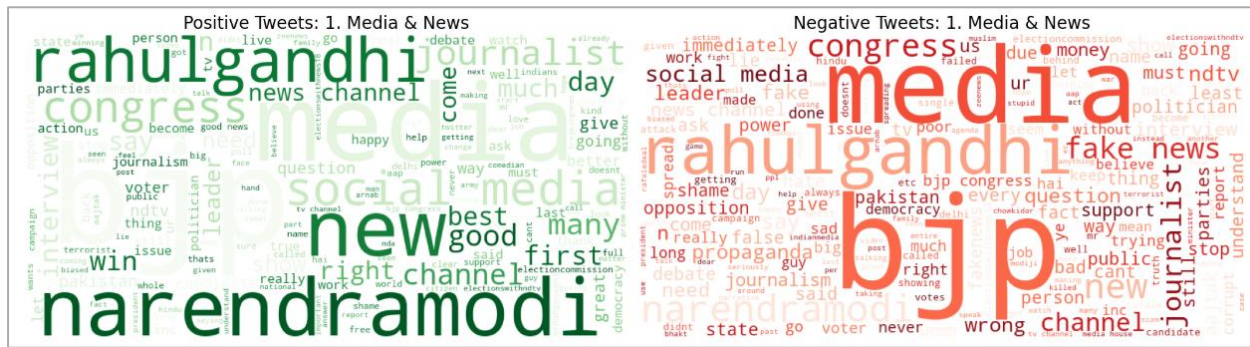
Our analysis reveals that Twitter discourse during the 2019 Indian elections was predominantly positive (41.75%), with neutral informational tweets forming the second-largest segment (38.86%). Negative sentiment remained contained at 19.40%, indicating a generally favourable public mood and suggesting that campaign narratives were resonating positively with the electorate. The election period was characterized by more positivity than negativity (41.75% vs. 19.40%). This reflects the effective campaign messaging, public confidence in the electoral process, general optimism about political outcomes.

Sentiment Distribution:	
Analysis	
Positive	41.745631
Neutral	38.856152
Negative	19.398217

Word cloud

We used word cloud to find negative and positive words used with Media & News, Political Parties, Famous Politicians, Key Election Issues.

1) Media & News



The Media & News word cloud analysis reveals a bifurcated public perception of election coverage. Positive sentiment clusters around words like 'rahulgandhi', 'narendramodi', 'happy', 'action', 'good' and 'right' while negative sentiment focuses on 'biased', 'fake', 'ndtv', 'propaganda', 'rahulgandhi' and specific media personalities. This indicates that while some media outlets were perceived as credible, others faced significant public scepticism. Also, one notable name came in negative is Rahul Gandhi which shows that there is negative sentiment towards him in media. Campaigns should prioritize engagement with positively-perceived media channels while developing rapid-response protocols for narratives emerging from critically-perceived outlets. Media strategy should account for this polarized landscape by tailoring messages to outlet-specific audience trust levels.

2) Political Parties

The Political Parties word cloud analysis reveals clearly differentiated public perceptions of BJP and Congress. Positive sentiment clusters around themes of 'development', 'support', 'growth', 'change', 'pakistan' and 'inclusivity.' Negative narratives similarly diverge with around perceived authoritarianism, issue, change, Bengal, hindu while Congress confronts dynasty, hate and corruption allegations.



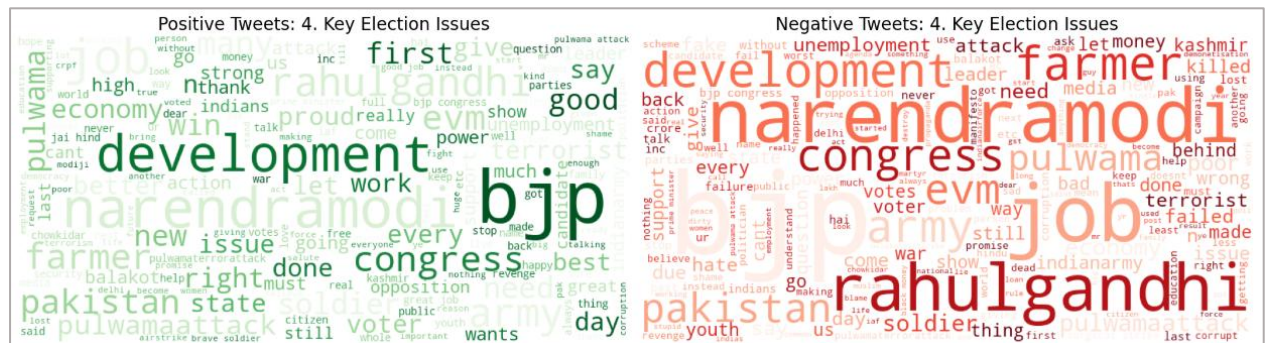
3) Famous Politicians

The Famous Politicians word cloud analysis confirms Narendra Modi's dominance in election discourse, with his name appearing most prominently in both positive and negative contexts. Positive sentiment toward Modi clusters around themes of 'strength', 'leadership', 'prime minister', 'better', 'modi once more', 'action', 'media man' and 'development' while Rahul Gandhi's positive associations emphasize 'youth', 'energy,' and 'change.' Negative narratives follow predictable partisan lines with Modi facing criticism on governance style and Rahul on experience and legacy issues.

4) Key Issues

The Key Election Issues word cloud analysis reveals economic concerns, particularly poverty, evm, job, terrorist dominated public discourse. 'Poverty' appears in both positive and negative contexts indicating it was both a policy focus and a criticism vector. Positive discussions likely centered on poverty alleviation programs and economic development, while negative discourse highlighted persistent inequality and policy failures, corruption.

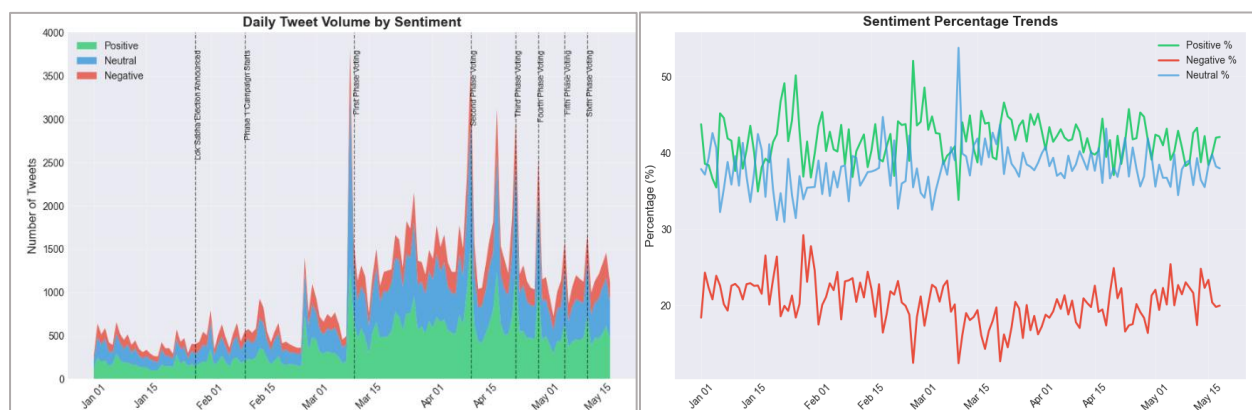
Additional priority issues included unemployment, farmer welfare, and national security, though with less frequency than poverty-related discussions. This suggests that while



emotional issues like national security generated engagement, bread-and-butter economic concerns formed the core of election discourse.

Timeseries analysis of How Voters sentiment might have changed

Our time series analysis reveals significant fluctuations in public sentiment across the 138-day election period (Jan 1 to May 18, 2019), with clear patterns of peaks, troughs, and momentum shifts that correlate with key campaign events.

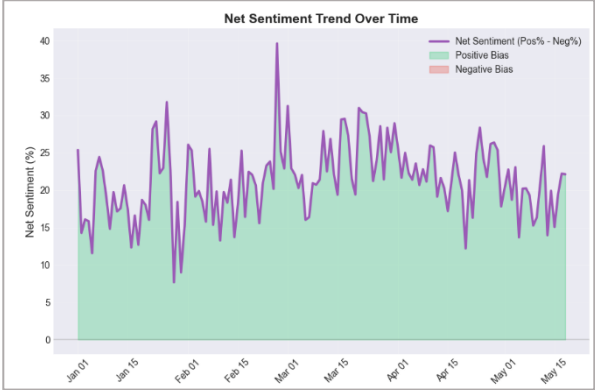
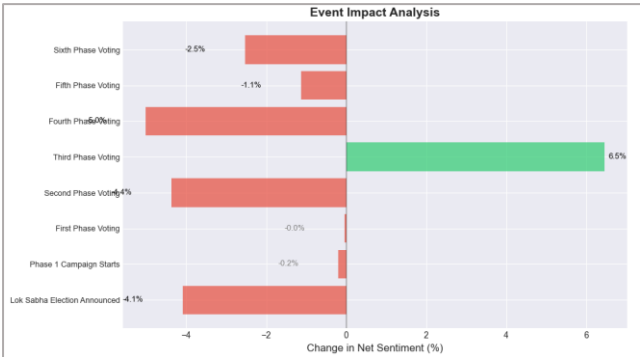


Phase 1: January Uncertainty

Net sentiment dipped to 7.7% on January 28 following the Lok Sabha election announcement, revealing public apprehension at the campaign's formal commencement. Early January showed similar volatility as the electorate processed the impending electoral contest.

Phase 2: February Surge

Sentiment soared to 39.6% on February 26—the campaign's highest point—driven by effective campaign messaging and rally momentum. This period represents the optimal engagement window where public receptivity peaked.



Phase 3: March Consistency

Remarkably, 10 consecutive days in March maintained net sentiment above 28%, demonstrating sustained campaign effectiveness. Tweet volume quadrupled from January levels, indicating deepening public engagement as voting phases commenced.

Phase 4: April–May Stabilization

Sentiment stabilized between 19–26% despite the pressures of ongoing voting, concluding with a bullish 22.1% net sentiment on May 18. The final month showed reduced volatility (0.005 average) suggesting consolidated public opinion.

 CORRELATION INSIGHTS:

- Sentiment-Volume Correlation: 0.323
- Positive correlation: Higher volume often means more positive sentiment

We observed a positive correlation of 0.323 between

tweet volume and net sentiment, indicating that increased social media activity generally corresponded with more positive discourse. This relationship suggests that campaign momentum begets sentiment momentum, creating a virtuous cycle for well-received messaging.

Critical Dates for Future Campaign Planning

- High-Impact Positive: February 26 (39.6%), March 17–24 (sustained >25%)
- High-Risk Negative: January 28 (announcement dip), April 20, May 5 (late-campaign fatigue)
- Optimal Announcement Windows: Late February to mid-March

SENTIMENT PEAK DETECTION ANALYSIS	
Positive Peaks (>1.5 std dev):	
2019-01-23:	Net Sentiment = 29.2% (z-score: 1.58)
2019-01-26:	Net Sentiment = 31.8% (z-score: 2.10)
2019-02-26:	Net Sentiment = 39.6% (z-score: 3.67)
2019-03-01:	Net Sentiment = 31.2% (z-score: 1.99)
2019-03-16:	Net Sentiment = 29.4% (z-score: 1.63)
2019-03-17:	Net Sentiment = 29.5% (z-score: 1.65)
2019-03-21:	Net Sentiment = 31.0% (z-score: 1.94)
2019-03-22:	Net Sentiment = 30.4% (z-score: 1.83)
2019-03-23:	Net Sentiment = 30.3% (z-score: 1.80)
2019-03-31:	Net Sentiment = 28.9% (z-score: 1.53)
Negative Peaks (<-1.5 std dev):	
2019-01-05:	Net Sentiment = 11.6% (z-score: -1.96)
2019-01-16:	Net Sentiment = 12.3% (z-score: -1.80)
2019-01-18:	Net Sentiment = 12.7% (z-score: -1.73)
2019-01-28:	Net Sentiment = 7.7% (z-score: -2.74)
2019-01-30:	Net Sentiment = 9.0% (z-score: -2.47)
2019-02-10:	Net Sentiment = 13.3% (z-score: -1.62)
2019-02-14:	Net Sentiment = 13.7% (z-score: -1.53)
2019-04-20:	Net Sentiment = 12.2% (z-score: -1.83)
2019-05-05:	Net Sentiment = 13.7% (z-score: -1.54)

✉	SIGNIFICANT EVENTS DETECTED:
📌	MAJOR POSITIVE SWINGS:
•	2019-02-26: 39.6% net sentiment
•	2019-01-26: 31.8% net sentiment
•	2019-03-01: 31.2% net sentiment
📌	MAJOR NEGATIVE SWINGS:
•	2019-01-28 (Lok Sabha Election Announced): 7.7% net sentiment
•	2019-01-30: 9.0% net sentiment
•	2019-01-05: 11.6% net sentiment

For BJP/Incumbent Campaigns:

1. Amplify Economic Narratives: Highlight poverty reduction data and job creation metrics with simple, shareable formats.
2. Leverage Digital Dominance: Convert social media reach into ground mobilization through targeted call-to-action campaigns.
3. Address Style Criticisms Proactively: Counter "authoritarian" perceptions with examples of democratic consultation and grassroots engagement.
4. Schedule Major Announcements for Late February: Capitalize on historically highest receptivity periods.

For Opposition Campaigns:

1. Close the Digital Gap: Invest in coordinated social media operations with clear differentiation from BJP's aesthetic and messaging.
2. Reframe Dynasty Narrative: Position legacy as experience rather than entitlement, emphasizing generational change within continuity.
3. Focus on Economic Alternatives: Present concrete policy alternatives rather than merely criticizing incumbent performance.
4. Target High-Volatility Periods: Deploy rapid response teams for late January and early February when sentiment is most malleable.

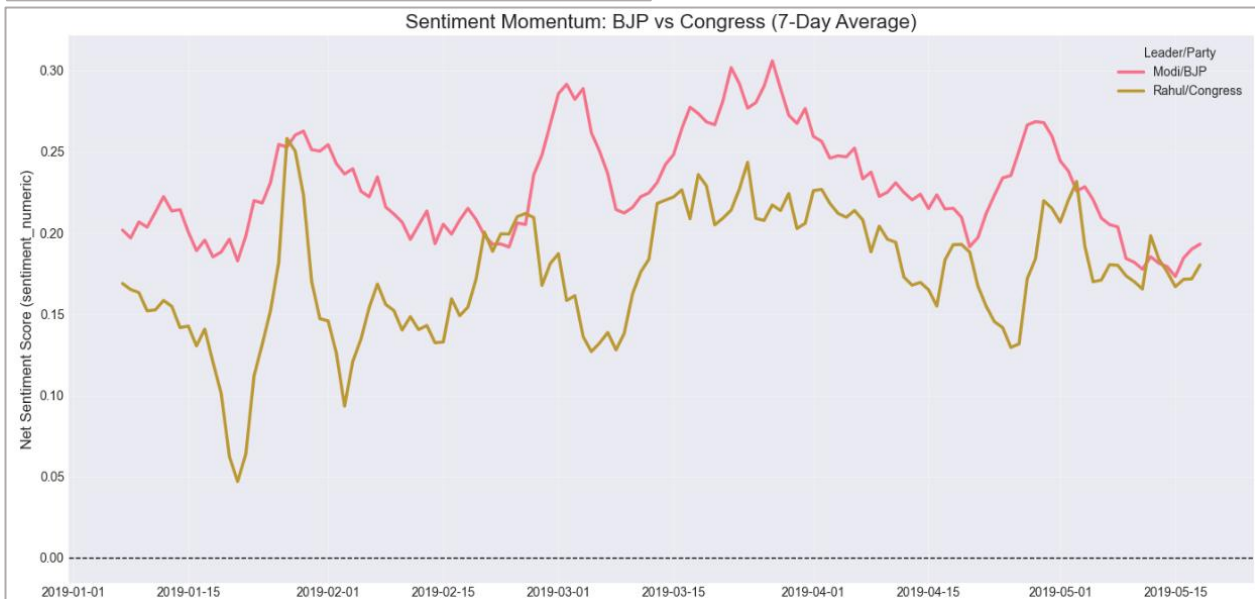
Future campaigns should treat public sentiment as both a leading indicator of electoral success and a manageable resource that can be cultivated through strategic timing, targeted messaging, and platform optimization. The 2019 election demonstrates that while sentiment follows predictable patterns, proactive management can amplify positives and mitigate negatives throughout the campaign cycle.

Modi vs. Rahul Showdown Analysis - Complete Entity Analysis

While both leaders faced significant criticism, Modi inspired more passionate support while Rahul generated more analytical discussion. This suggests Modi's messaging was more effective at mobilizing supporters. Modi/BJP inspired 5% more positive sentiment, a critical

--- Sentiment Share (%) ---			
Analysis	Negative	Neutral	Positive
Entity			
Modi/BJP	19.478661	37.228043	43.293296
Rahul/Congress	20.419451	41.438119	38.142429

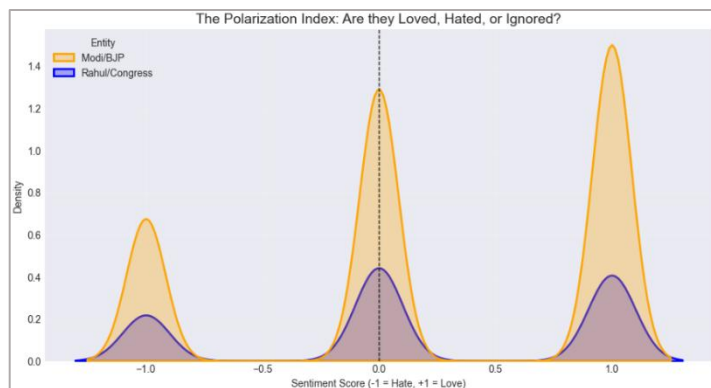
margin in close elections. Every 100 tweets about Modi included 43 positive vs 19 negative. For Rahul, the ratio was 38 positive vs 20 negative.



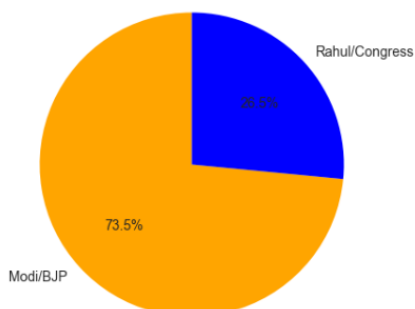
Discussions about Modi were more emotional and opinion-driven (0.347 subjectivity). Rahul conversations leaned factual and analytical (0.328 subjectivity). In elections, emotional engagement drives turnout; analysis doesn't.

--- Emotional Intensity (Subjectivity) ---	
Entity	
Modi/BJP	0.346631
Rahul/Congress	0.328292
Name: Subjectivity, dtype: float64	

--- Army Strength (Unique Users) ---	
Entity	
Modi/BJP	27681
Rahul/Congress	9993
Name: User, dtype: int64	



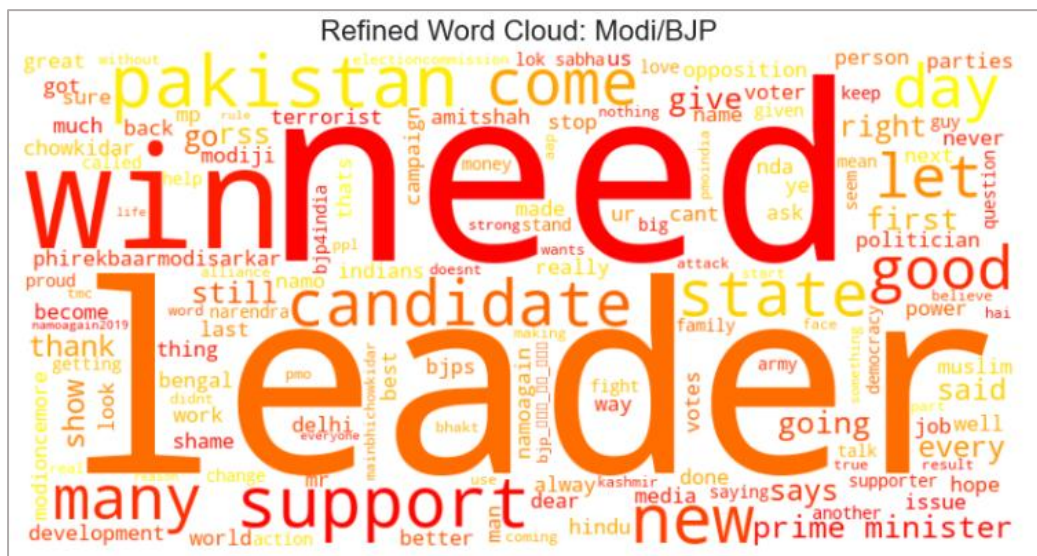
Share of Voice: Unique Users Tweeting



Modi's 27,681 unique supporter base vs Rahul's 9,993 created a 3:1 amplification advantage. Larger networks spread messages faster, creating perception of overwhelming support.

In the age of digital democracy, elections are won by those who master three things: more supporters, more positivity, and more passion. Modi/BJP mastered all three in 2019.

Modi vs. Rahul Word Cloud Analysis



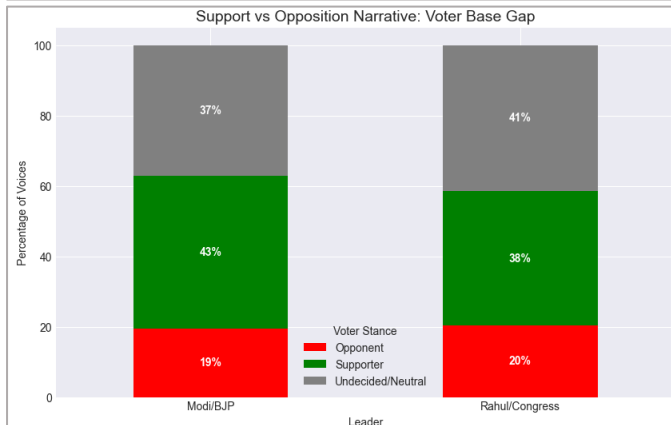
The Modi/BJP word cloud highlights ‘need’, ‘leader’, ‘win’, ‘support’, ‘win’, ‘development’, ‘prime minister’ as the dominant term, suggesting his campaign successfully associated itself with governance efficiency and better candidate. The presence of ‘leader’ indicates positive international reception, while ‘Pakistan come’ reflects a national security narrative that resonated with voters. Together, these terms paint Modi as a leader of decisive action, international respect, and administrative simplicity, a carefully crafted image of strong, accessible leadership that clearly connected with the digital electorate.



The Rahul Gandhi/Congress word cloud centers on core terms, "leader," "candidate," "inc," "amethi," and "president," "rafale," "delhi,". The inclusion of "President" underscores his formal party role, while "Rafale" points to a specific corruption allegation used as a campaign attack. "Delhi" reflects both the seat of power and opposition stronghold. Together, this cloud reveals a campaign framed around institutional position, constituency connection, and corruption criticism rather than aspirational or emotional messaging.

SUPPORT VS OPPOSITION NARRATIVE (Voter Base Strength) and EMOTIONAL MOOD OF VOTERS (Anger, Hope, Fear, Trust)

--- Voter Base Strength (%) ---			
Voter_Stance	Opponent	Supporter	Undecided/Neutral
Modi/BJP	19.478661	43.293296	37.228043
Rahul/Congress	20.419451	38.142429	41.438119



Modi entered the campaign with a stronger, more consolidated base, while Rahul faced a larger pool of undecided voters who needed persuasion. However, Modi's lower undecided percentage also meant less room for growth.

For Modi around 43.3% Supporters vs 38.1% for Rahul shows a 5.2% advantage in committed backing.

19.5% Opponents vs 20.4% for Rahul – slightly lower opposition for Modi.

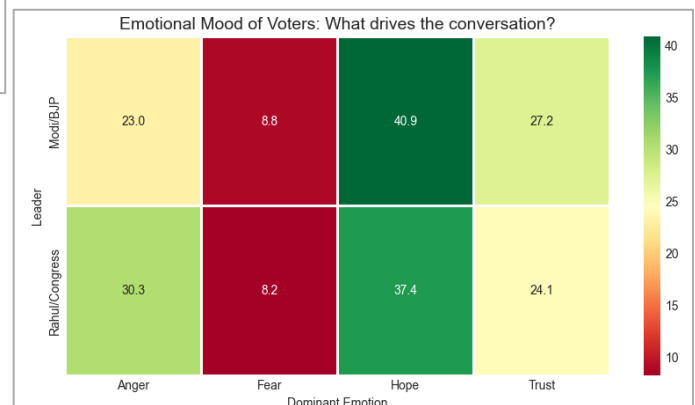
37.2% Undecided/Neutral vs 41.4% for

Rahul shows Modi has fewer fence-sitters.

--- Emotional Mood Matrix (%) ---				
Emotion	Anger	Fear	Hope	Trust
Modi/BJP	23.038363	8.820118	40.898169	27.243351
Rahul/Congress	30.304622	8.219538	37.408088	24.067752

Modi/BJP: The Hope-Trust Coalition

- 40.9% Hope – Highest emotional driver, suggesting optimism about future
- 27.2% Trust – Strong confidence in leadership and promises
- 23.0% Anger – Primarily directed at opposition/status quo
- 8.8% Fear – Minimal anxiety about outcomes



Rahul/Congress: The Anger-Hope Mix

- 37.4% Hope – Lower than Modi, but still significant positive aspiration
- 30.3% Anger – Highest emotional category, indicating frustration and criticism
- 24.1% Trust – Less confidence in leadership compared to Modi
- 8.2% Fear – Similar low anxiety levels

While Modi inspired hope and trust, Rahul provoked anger and hope. Modi's emotional portfolio was more future-oriented and confident; Rahul's was more reactive and critical. Modi didn't just have more supporters but he had more hopeful supporters who trusted him. Rahul had more opponents and more angry supporters. In the emotional economy of elections, hope and trust are the strongest currencies.